(Discrete Mathematics)　　22BCON1068

Assignment - 01

DATE: 16/9/23
PAGE NO.: 01

• PART - A ($5 \times 2 = 10$)

Q A1. if $S = \{1,2,3,4\}$ and $T = \{4,5,6,7\}$. What is $S \Delta T$ (S is symmetric diffⁿ of T) ?

Solⁿ.

$$S = \{1,2,3,4\}$$
$$T = \{4,5,6,7\}$$
$$S - T = \{1,2,3\}, \quad T - S = \{5,6,7\}$$
$$\therefore S \Delta T = \{1,2,3,5,6,7\}$$

Q A2 Find the domain and range of the function $f(x) = \frac{x}{1-x}$.

Solⁿ

$$f(x) = \frac{x}{1-x} \Rightarrow 1-x \neq 0 \rightarrow x \neq 1$$
$$x \in (-\infty, \infty) - \{1\} \leftarrow \text{Domain}$$
$$y = \frac{x}{1-x} \Rightarrow y - xy = x$$
$$\Rightarrow xy + x = y$$
$$\Rightarrow x(1+y) = y$$
$$x = \frac{y}{1+y}$$
$$1+y \neq 0 \Rightarrow y \neq -1$$
$$\boxed{y \in R - \{-1\} \Leftarrow \text{Range.}}$$

Q3. Let S be the set of letters in the English alphabet. What is the cardinality of S ?

Solⁿ.

Cardinality of finite set is $n(A) = a$ where, a is the no. of element of set A

$$\boxed{\therefore \text{Cardinality of } S = 26}$$

Dec 26, 2023, 16:32

Q4. What is the power set of the empty set? What is the power set of set $\{\phi\}$?

Ans

The power set of the empty set is $\{\phi\}$
The power set of the set $\{\phi\} = \{\phi, \{\phi\}\}$

Q5. What is the cartesian Product of $A = \{1,2\}$ and $B = \{a,b,c\}$?

Sol^n. $A = \{1,2\}$, $B = \{a,b,c\}$

$A \times B = \{(1,a), (1,b), (1,c), (2,a), (2,b), (2,c)\}$

• PART - B ($3 \times 7 = 21$)

Q1. Let f_1 and f_2 be functions from R to R such that $f_1(x) = x^2$ and $f_2(x) = x - x^2$. What are the functions $f_1 + f_2$ and $f_1 f_2$?

Sol^n.

$f_1 + f_2 = f_1(x) + f_2(x) = \cancel{x^2} + x - \cancel{x^2}$

$\boxed{f_1 + f_2 = x}$

$f_1 f_2 = f_1(x) \cdot f_2(x) = (x^2) \cdot (x - x^2)$

$\boxed{f_1 f_2 = x^3 - x^4}$

Q2. Let f be the function from R to R with $f(x) = x^2$. is f invertible.

Sol^n To check $f(x)$ is one-one : for $x_1, x_2 \in R$

$f(x_1) = f(x_2)$

$x_1^2 = x_2^2$

$x_1 = \pm x_2$

$\therefore$ The f^n is not one - one.

$\rightarrow$ To check $f(x)$ is onto :-

Let $y \in R$ such that

$y = f(x) \Rightarrow y = x^2$

$\sqrt{y} = x$

$y \geq 0 \Rightarrow f^n$ is not onto Function.

$\therefore$ f^n f is not invertible since f is neither one - one nor onto.

Q3. Let f and g be the fun. from the set of integers to the set of integers defined by $f(x) = 2x+3$ and $g(x) = 3x+2$. What is the composition of f and g? What is the composition of g and f?

Sol^n.

$$(f \circ g)(x) = f[g(x)] = 2[g(x)] + 3$$
$$= 2(3x+2) + 3$$
$$= 6x + 4 + 3$$
$$\boxed{(f \circ g)(x) = 6x+7}$$

$$(g \circ f)(x) = g[f(x)] = \cancel{8}$$
$$= 3[f(x)] + 2$$
$$= 3(2x+3) + 2$$
$$= 6x + 9 + 2$$
$$\boxed{(g \circ f)(x) = 6x+11}$$

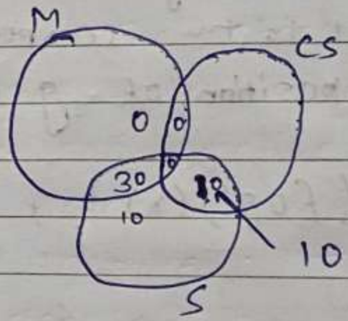
As

• PART-C (3×11=33)

Q1. in a group of 90 student, each of whom has taken at least mathematics or computer science or statistics, it was found 40 students having mathematics, 50 student having computer science and 60 student having statistics. 10 students having mathematics as well as computer science, 40 students have mathematics as well as statistics and 10 student have all the three subject. Find the no. of students who have computer science and statistics but not mathematics.

Sol^n.

$n(m) = 40$
$n(cs) = 50$
$n(s) = 60$
$n(m \cup cs) = 20$
$n(M \cup S) = 40$
$n(M \cap S \cap CS) = 10$
$n(S \cup CS) = ?$



$n(M \cup CS \cup S) = n(m) + n(cs) + n(S) - n(m \cup cs)$
$- n(M \cup s) - n(Cs \cup S) + n(M \cap Cs \cap S)$

$90 = 40 + 60 + 50 - \cancel{10} - 40 - n(S \cup Cs) + \cancel{10}$

$90 = 110 - n(S \cup Cs)$

$n(S \cup Cs) = 20$

~~The~~ now, $n(S \cup Cs) - n(M \cup Cs)$
$= 20 - 10$
$= 10$

no. of students who have CS and statistics but not math $= 10$.

Ans

DATE : / /

PAGE NO. : 05

Q2. in a language survey of student of a college in Chennai, it was fount that 900 student know English, 1200 Student know Tamil and 750 student know Hindi. 450 student know English and Tamil 300 student know English and Hindi while 200 student know Tamil and Hindi. 150 students know all three languages. Find how many students know -

(a) Hindi only.

(b) At Least one Language

(c) English and one but not both Tamil and Hindi.

(d) At least Two Languages.

Sol.

$n(Eng) = 900$

$n(Tamil) = 1200$

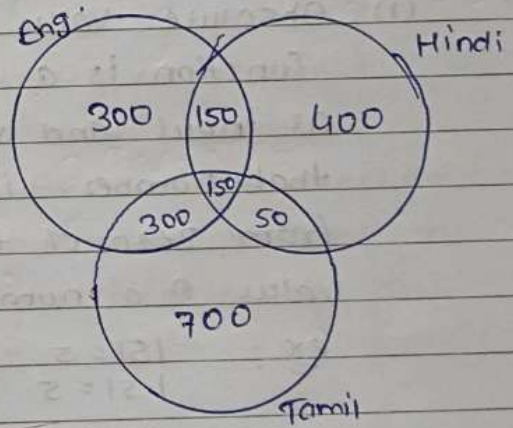
$n(Hindi) = 750$

$n(Eng. \cap Tamil) = 450$

$n(Eng. \cap Hindi) = 300$

$n(Tamil \cap Hindi) = 200$

$n(Eng. \cap Hindi \cap Tamil) = 150$



(a) Hindi only - 400

(b) At least one languag =

$900 + 1200 + 750 - 450 - 300 - 200 + 150 = 2050$ As

(c) English and one but not tamil and Hindi =

$300 + 150 = 450$

(d) At least Two Languages = $350 + 150 + 150 + 050$

$= 700$

Q3. Define Absolute value function, floor function, ceiling function, mod function and Div function with an example.

Sol.

(i) Absolute value function :- The absolute value function is a function that takes a real number as input and returns the non-negative value of that number. in other words, it takes the distance from zero of the input number. The absolute value of a number is denoted by |x|.

ex : $|5| = 5$
$|-5| = 5$
$|0| = 0$

(ii) Floor function :- The floor function is a function that takes a real number as input and returns the greatest integer less than or equal to that number.

ex :- $3.5 = 3$
$-3.5 = -4$

(iii) Ceiling function :- The ceiling function is a function that takes a real number as input and returns the least integer greater than or equal to that number.

ex : $3.5 = 4$
$-3.5 = -3$

22BCON1068

DATE: / /
PAGE NO.: 07

(iv) Mod Function :- The mod function is a function that takes two integers as input and returns the remainder of the first integer divided by the second integer.

The modulus function $f(x)$ of x is defined as;

$f(x) = |x|$ or $y = |x|$

where $f: R \to R$ and $x \in R$.

and $|x|$ states modulus or mod of x.

ex:

if $x = -5$, then $y = f(x) = -(-5) = 5$, $x < 0$.

if $x = 10$, then $y = f(x) = 10$, $x > 0$.

if $x = 0$, then $y = f(x) = 0$, $x = 0$.

(v) Div Function :- The Div fun. is a function that takes two integers as input and returns the quotient of the first integer divided by the second integer.

Eg: div(12,5) = Q = 3, R = 2

12

30/12